

Justice & Global Health Research

COLLEGE117

APRIL 22, 2026

Overview

Monday: Research Infrastructure

- Global Funding & Research

Wednesday: Research Disparities

Historical disparities in biomedical research

- Justice frameworks
  - Cahill: Justice & Biotech
  - Pratt: Social Justice for Global Health Research

How is diversity relevant to biomedical/health research?

Diversity for research – why need diversity?

- Diversity is fundamental to rigorous biomedical research
- In human research – need for understanding or having the research reflect variation within populations
  - Social benefit for all
    - Research population needs to reflect target population

Conceptualizing diversity

- Consider the research question
- Many kinds of diversity needed (e.g. SES, age, geographic)
  - Racial diversity sometimes used as proxy for other relevant traits
  - Participation in research may also improve trust and other values
  - Achieving appropriate population diversity in research while avoiding racial essentialism

Construction of Medical Knowledge



Funding



Research



Methods



Participants/Data

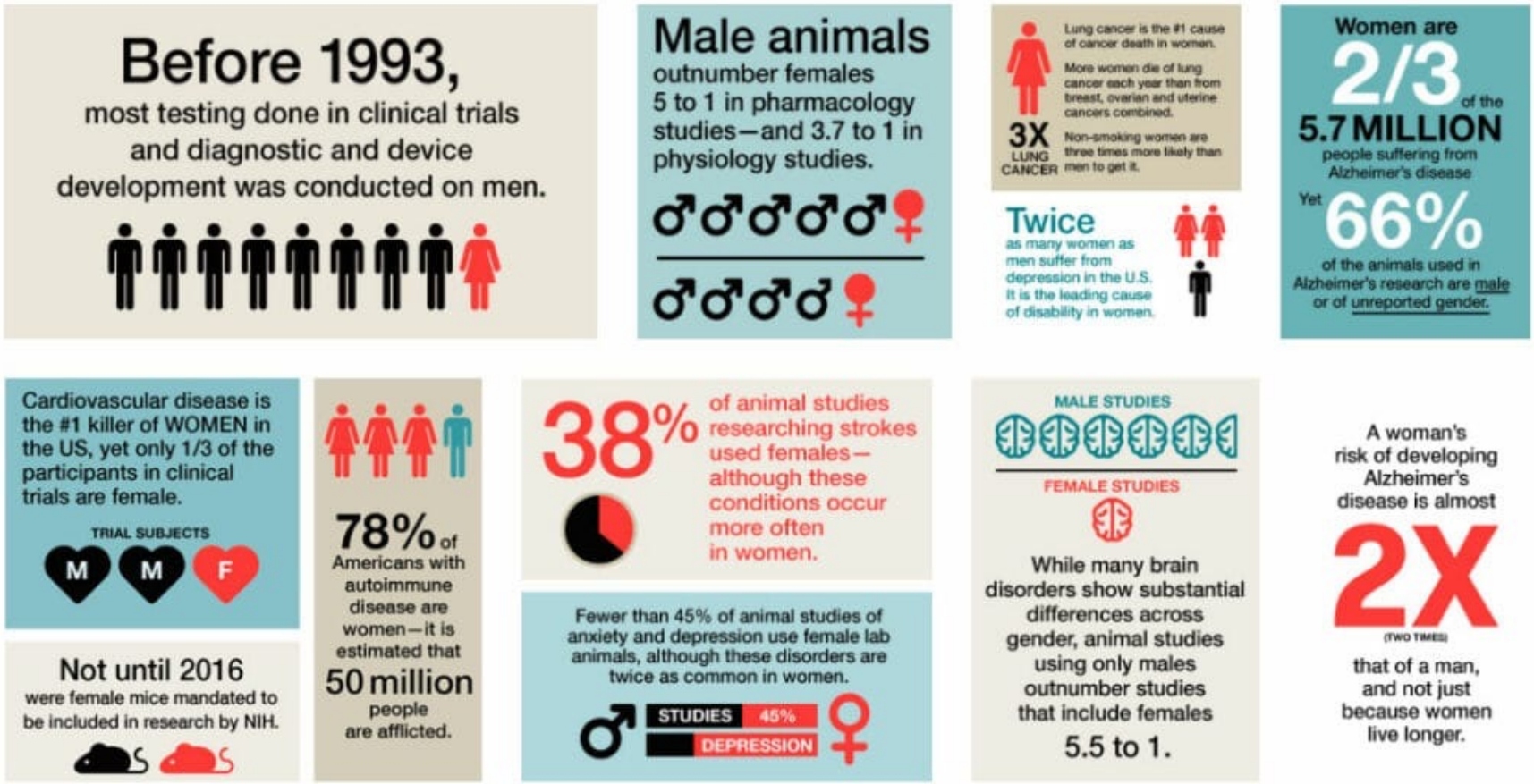


Publication



Implementation:
Clinic & Education

Disparities in Alzheimer's research



News Feature | Published: 14 March 2025

Women twice as likely to develop Alzheimer's disease as men — but scientists do not know why

Sofia Moutinho

Nature Medicine **31**, 704–707 (2025) | Cite this article

35k Accesses | **230** Altmetric | Metrics

Changes to the female brain during menopause, together with social and cultural factors, might affect disease risk.

ALZHEIMER'S ASSOCIATION | Alzheimer's & Dementia®
THE JOURNAL OF THE ALZHEIMER'S ASSOCIATION

RESEARCH ARTICLE | Open Access

Early life exposure to structural sexism and late-life memory trajectories among black and white women and men in the United States

Justina F. Avila-Rieger, Paris B. Adkins-Jackson, Tanisha G. Hill-Jarrett, Whitney R. Robinson, Katherine M. Keyes, Nicole Schupf, Adam M. Brickman, Richard P. Mayeux, Jennifer J. Manly

First published: 18 December 2024 | https://doi.org/10.1002/alz.14410 | Citations: 1

Lack of racial diversity in biomedical research subjects (Oh et al., 2015)

- Racial and ethnic minorities make up nearly 40% of the United States population
- 80-90% of clinical trial participants are white non-Hispanic
- Ignoring the racial/ethnic diversity of the US population is a missed scientific opportunity to fully understand the factors that lead to disease or health.

Minority Representation in Biomedical Research



The Problem with Pulse Oximeters: A Long History of Racial Bias

Racial bias in pulse oximetry has been documented for decades—why hasn't more been done to fix it?

Published **July 08, 2024**

By **Annalies Winny**, Nicole Jurmo

COVID-19 | HEALTH EQUITY | MEDICINE | TECHNOLOGY AND INNOVATION



Medical knowledge --> Education

A medical student couldn't find how symptoms look on darker skin. He decided to publish a book about it.

By Sydney Page
July 22, 2020



Medical education

- 2016: 50% of white medical students believed that Black people feel less pain and/or have thicker skin than white people
- These types of beliefs lead to undertreatment of pain in Black patients in clinical care

RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES | FREE ACCESS

Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites

Kelly M. Hoffman, Sophie Trawalter, Jordan R. Axt, and M. Norman Oliver Authors Info & Affiliations

April 4, 2016 | 113 (16) 4296-4301 | https://doi.org/10.1073/pnas.1516047113

MEDICINE AND SOCIETY

Misrepresenting Race — The Role of Medical Schools in Propagating Physician Bias

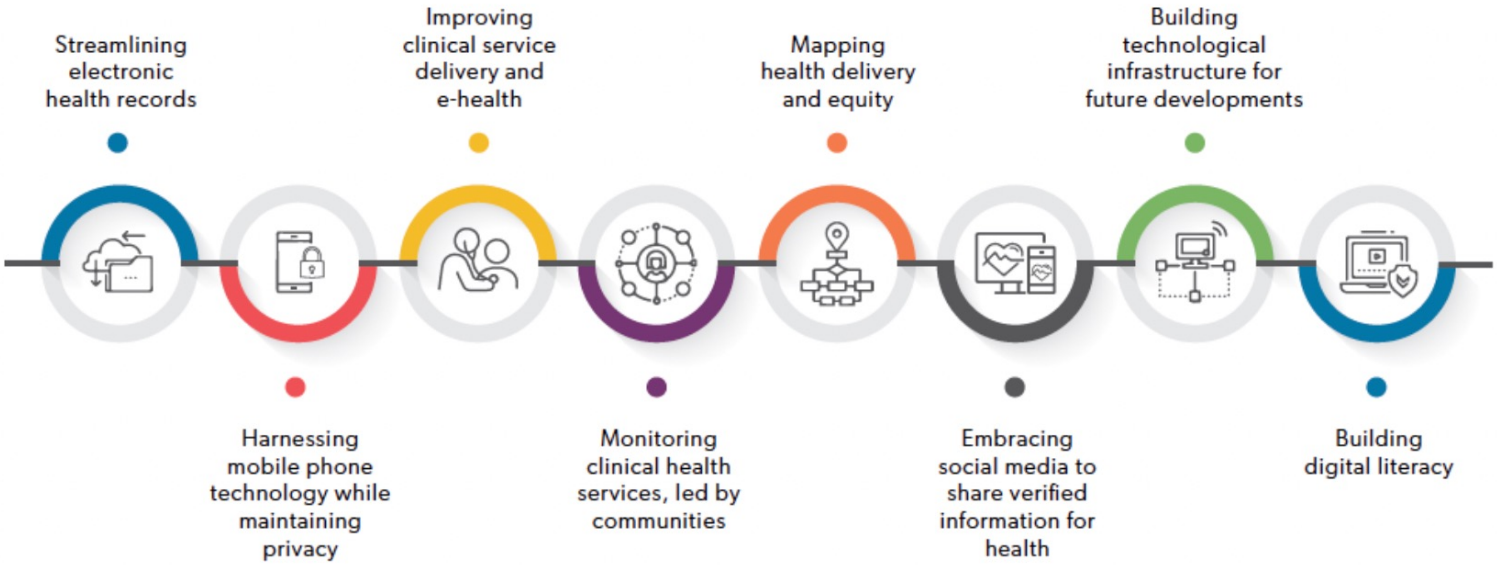
Christina Amutah, B.A., Kaliya Greenidge, Adjoa Mante, A.B., Michelle Munyikwa, Ph.D., Sanjna L. Surya, B.A., Eve Higginbotham, M.D., David S. Jones, M.D., Ph.D., Risa Lavizzo-Mourey, M.D., M.B.A., Dorothy Roberts, J.D., Jennifer Tsai, M.D., M.Ed., and Jaya Aysola, M.D., D.T.M.H., M.P.H.

Article Figures/Media Metrics March 4, 2021
N Engl J Med 2021; 384:872-878

Neurotech and inclusive tech for EEGs (3:55)



HOW HEALTH TECHNOLOGY CAN ADVANCE GLOBAL HEALTH EQUITY



PRINCIPLES THAT SHOULD GUIDE TECHNOLOGY:

1. Strengthen public health systems
2. Put people at the center of innovations
3. Ensure equity in access and protect privacy and individual rights

FRIENDS OF THE GLOBAL FIGHT | AGAINST AIDS, TUBERCULOSIS AND MALARIA

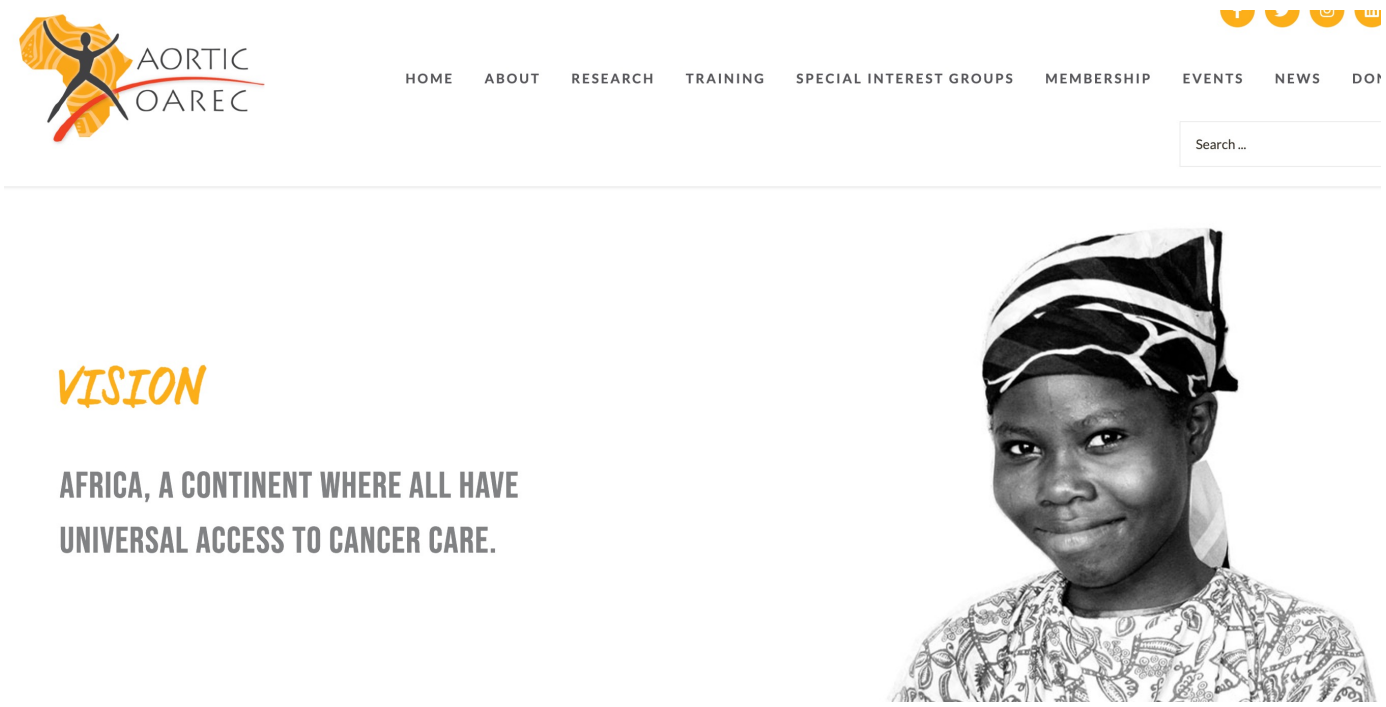
Equity in international health research collaborations in Africa: Perceptions and expectations of African researchers

Nchangwi Syntia Munung, Bongani M. Mayosi, Jantina de Vries

Published: October 16, 2017 • https://doi.org/10.1371/journal.pone.0186237

Capacity building
Sustainability
Avoiding exploitation of researchers

African Organisation for Research and Training in Cancer



Transformation global health research (Pratt)

- Shifting control away from global North
  - Global health research funding, education, conduct & publishing
- Making knowledge from global South visible and valued

Global Health Research & Health disparities

Global health research should generate knowledge to reduce health disparities (Pratt)

Lack of national public health and biomedical data from research in Africa poses challenges in addressing the health burden (Mulder, et al, 2021)

Five core dimensions of social justice (Pratt)

- Power
- Recognition
- Harmony
- Inclusion
- Well-being

Pratt: core dimensions for just global health research

Power: " reducing unfair power dynamics between humans"

Well-being: "ethical significance of the functioning and flourishing of humans and nonhumans and finds harm—injustice—where that potential is limited and deprivation occurs"

Pratt: core dimensions of just global health research

Harmony: "reflects the ethical significance of relationships of shared identity, care, and solidarity between humans and finds harm—injustice—in forces that interfere with those relations"

Recognition: "morally necessary because misrecognition causes status injury (i.e., disrespect or erasure) to social groups, cultures, and/or their ways of knowing"

What is an example of what is needed for just inclusion of LMIC country researchers in global health research?

Pratt: core dimensions

- Inclusion: “calls for meaningful participation in decision-making processes by all those affected, particularly those affected who are considered socially marginalized or disadvantaged”

Building research capacity in South Africa: Dr. Olivia Matshabane
5:30



News

Dr. Olivia Matshabane receives one of the 2025 IBRO Rising Stars Awards

JUNE 2025




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Summary

- This week:
  - International research infrastructure impacts the development of health technology and distribution of health resources
- Next week: What factors impact the development and implementation of just genomic and precision medicine and in what ways?